List of revisions made to "On Vassiliev Invariants and Weight Systems of Classical and Welded Knots"

There are two reports for this thesis, a report by Audoux and a report by an anonymous reviewer.

The report by Audoux contains issues to be corrected / suggestions, of which six are identified as major, and twenty-nine are identified as minor. The report also lists some possible typos for correction. We will reference these issues as Audoux.M1 through Audoux.M6 for the major issues, and Audoux.m1 through Audoux.m29 for the minor issues.

The report by Raynor contains two general required revisions in Section 2.1, a list of required revisions by section of the thesis, and a list of recommendations for future publication of the thesis, also by subsection of the thesis. We refer to the revisions in this report by the sections of the report, then by the sections of the thesis which they address.

Table 1: List of revisions made based on corrections, suggestions or comments in review reports

Position in submitted thesis	**Position in revised thesis**	**Modification made**	**Relevant issue(s) in report(s)**
Paragraph after Remark 1.1.2	Paragraph after Remark 1.1.2	Be explicit about use of phrases "positive and negative crossings" when only dealing with embeddings in $\mathbb{R}^3$.	Audoux.m1
-	Definition 1.1.3	Define knot invariant and provide reference for definition of ambient isotopy. Assert that everything in the text is done over $\mathbb{Q}$.	Audoux.M1
Start of Section 2.2	Start of Section 2.2	Small changes to wording to make meaning clearer	Audoux.M2
-	Definition 2.2.1	Give 'tensor' its own environment-level definition	Audoux.M2
-	Proposition 2.2.2 and its proof	Provide details for the isomorphism between multilinear maps and elements of the space $X_1^* \otimes \cdots \otimes X_n^* \otimes Y_1 \otimes \cdots \otimes Y_m$	Audoux.M2
First diagram of Section 2.2, diagrams of section 3.3	First diagram of Section 2.2, diagrams of section 3.3	Remove dual notation (e.g. x^*, y^*) from string diagrams, as the top-to-bottom convention instead suffices to determine whether they are inputs or outputs	Audoux.M2
Sections 2.2, 2.3	Sections 2.2, 2.3	Remove arrows from string diagrams where they're unnecessary, adressing confusion	Audoux.M2
-	Definition 2.2.4	Definition environment for metric Lie algebra	Audoux.M2
Question 2.2.7 Question 1.5.12	Remark 2.2.11 Remark 1.5.12	Change environment type of open questions from **Question** to **Remark**. This addresses the point in Raynor.2.1 about the use of the Question environment, as now (using the submitted version numbering) 1.2.2 and 2.2.9 describe natural questions to explore and pre-empt what follows, and remain as Question environments. Open problems which are given as curiousities and are not further explored in the thesis take the form of Remark environments	Raynor.2.1

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Table 1: List of revisions made based on corrections, suggestions or comments in review reports (Continued)

Third last paragraph of Section 2.3	Third last paragraph of Section 2.3	Emphasise that the theory of Section 2.3 leaves Question 2.2.9 (submitted numbering) an open question. The questions still explores and pre-empts Section 2.3, so remains as a Question environment.	Raynor.2.1
Paragraphs preceeding and proceeding Definition 1.2.3	Paragraph preceeding and proceeding Definition 1.2.3	Emphasise that the ways in which Definition 1.2.3 will answer Questions 1.2.2 will be explained in the current section.	Audoux.m2
Defintion 1.2.4	Definition 1.2.4	Include a proper definition of the signs of a singularity and state its equivalence to defining a coorientation on the $(m+1)$st stratum	Audoux.m3
Paragraphs preceeding Definition 1.2.5	Paragraphs preceeding Definition 1.2.5	Emphasise that the equation is true for any path	Audoux.m4
First paragraph of Section 1.3	First paragraph of Section 1.3	Remove unclear reference to linearisation/projectivisation.	Audoux.m6
First paragraph of Section 1.3	First paragraph of Section 1.3	Remove claim that Vassiliev invariants detect 3_1 implies that f_{3_1} is power series Vassiliev	Audoux.m7
	Before Definition 1.4.8	Specify meaning of “finite-type”	Audoux.m8
Remark 1.5.5	Remark 1.5.5	Remove typo to address apparent self-referencial definition of universal Vassiliev invariant	Audoux.m9
-	Proposition 2.2.5 and proof	Make clear the proposition status, and prove the relation on a metric Casimir coming from an invarint metric	Audoux.m10
Example 2.2.3	Example 2.2.8	Insert explicit computation of the Casimir element	Audoux.m12
Section 2.3	Section 2.3, Remark 2.3.1	Move parts of the suggested paragraph into the section introduction as suggested. As a result, this becomes the main point of the introduction paragraph and the part about historical terminology is moved into a new remark.	Audoux.m13
Theorem 2.3.8 to end of subsection	Theorem 2.3.9 to end of subsection	Small rephrasings to increase clarity. However to my knowledge it is not in the literature whether Lieberum’s order 15 example vanishes on Lie superalgebra weight systems.	Audoux m.14
Section 2.4	Section 2.4	State that $\mathfrak{g}_2$ and $\mathfrak{f}_4$ are the usual excpetional Lie algebras.	Audoux m.15
Example 2.5.3	Example 2.5.3	Fix typo, indeed vector space should be set	Audoux m.17
	Before Definition 2.5.4	Change ‘there is no natural notion of tensor product for operads’ to ‘the tensor product is not a natural operation for operads’, removing the ambiguous claim	Audoux m.18
Remark 2.5.6	Remark 2.5.6	Specify which functors are left- and right- adjoint	Audoux.m19
Definition 2.5.8	Definition 2.5.8	Add boldface text name “cyclic action condition” and rephrase to make obvious that the the relation is part of the definition of a cyclic operad	Audoux.m20

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Table 1: List of revisions made based on corrections, suggestions or comments in review reports (Continued)

Definition 2.5.9	Definition 2.5.9	Define the relevant symbols A and m in the same sentence	Audoux.m21
Paragraph before Definition 3.1.1 Definition 3.1.4	Paragraph before Definition 3.1.1 Definition 3.1.4	Accept the suggested wordings not to be factually misleading about welded knots	Audoux.m24
	Before Definition 3.1.3	Add suggested citation	Audoux.m25
Definition 3.2.5 Proposition 3.2.6 Section 3.2	Definition 3.2.5 - Section3.2	Define Jabobi diagrams as two-in-one-out by definition rather than proposition, remove all mention of the two-in-one-out rule	Audoux.M6, Audoux.m26
Section 3.3	Section 3.3	Clarify and reference citation that defines the semidirect product of $\mathfrak{g}$ on $\mathfrak{g}^*$ in the appropriate way	Audoux.m27
Definition 3.4.3	Definition 3.4.3	Specify as suggested that β is a generic bracket morphism, and so it has the same properties as α in Example 2.5.3	Audoux.m28
Entire text	Entire text	Fix typos	Both reviewers' identified typos except those identified in Table 2

Table 2: List of issues, comments or questions raised by the reviewers which are acknowledged but no modifications resulted

Relevant point(s) in report(s)	Explanation
Auduou.m4: why introduce the notation Q' when it's soon shown to be equal to $P = \partial Q$?	For the same reason we use the notation f and F when learning/proving the fundamental theorem of calculus: not to confuse the notation by a choice which would suggest the hypothesis. Also, to be consistent with the literature, specifically Willterton, which is cited.
Auduox.m5: Remark 1.2.18 is intriguing but uinnformative	Indeed, it is an open question to explain this, so there is nothing informative to say. However I feel that this intriguing point deserves to be stated.
Audoux.m11: why use the words covariant and contravariant?	They are standard enough, and could be useful for readers of less categorical backgrounds. Furthermore they are used a few more times in similar ways in the thesis.
Audoux.m20: Could the relation be pictured by ...	Such a picture would perhaps be enlightening but it would be factually incorrect.
Audoux.m29: "does it means that the universal welded weight system constructed in Section 3.4 is in some sense contained in $\mathfrak{gl}_{1\|1}$ in some way?"	Yes, in some way.
Audoux.typo19	No, the $\oplus 0$ is intentional and correct.